

CG Example

Ignacio Laguna

April 2026

1 Algorithm

Algorithm 1 Conjugate Gradient (FP32 baseline)

Require: SPD matrix A , RHS b , tolerance τ , max iterations $k_{\max}$

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1:  $x_0 \leftarrow 0, r_0 \leftarrow b - Ax_0, p_0 \leftarrow r_0$ 
2:  $\rho_0 \leftarrow r_0^T r_0, b_n \leftarrow \sqrt{b^T b}$ 
3: for  $k = 0, 1, \dots, k_{\max} - 1$  do
4:    $q_k \leftarrow Ap_k$ 
5:    $\alpha_k \leftarrow \rho_k / (p_k^T q_k)$ 
6:    $x_{k+1} \leftarrow x_k + \alpha_k p_k$ 
7:    $r_{k+1} \leftarrow r_k - \alpha_k q_k$ 
8:    $\rho_{k+1} \leftarrow r_{k+1}^T r_{k+1}$ 
9:   if  $\sqrt{\rho_{k+1}} / b_n < \tau$  then
10:    return  $x_{k+1}$ 
11:  end if
12:   $\beta_k \leftarrow \rho_{k+1} / \rho_k$ 
13:   $p_{k+1} \leftarrow r_{k+1} + \beta_k p_k$ 
14: end for
15: return  $x_{k_{\max}}$ 
```
